RYERSON UNIVERSITY

Department of Mechanical and Industrial Engineering

MIDTERM EXAM #1

Materials Science Fundamentals (MTL 200)

Name	ID	Signature
[redacted]	[redacted]	[redacted]

Instructions

1. This is a closed book examination. Class notes and textbooks are not permitted.
2. The student may use an electronic calculator (for calculations only; not for storage and access of formulas).
3. Please **use a pen only**. Pencil can be used only for drawings.
4. Exam duration: 1hour

Prepared by: Dr. H. Bougherara, W 2010

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MULTIPLE CHOICES (Answer all multiple choice questions) (5 points)

1- Which crystal system(s) listed below has (have) the following relationship for the unit cell edge lengths?
$a = b \neq c$

(a) Tetragonal
(b) Hexagonal
(c) Both A and B
(d) Triclinic

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2- Dislocations are line defects that produce :
(a) None
(b) Permanent (plastic) deformation
(c) Temporary (elastic) deformation
(d) Both C and B

3- (1) is used in diffraction experiments to determine (2).

(a) (1) X-ray (2) atomic weights
(b) (1) λ-ray (2) crystal structures
(c) (1) X-ray (2) crystal structures
(d) (1) λ-ray (2) atomic weights

4- Conditions for substitutional solid solution are

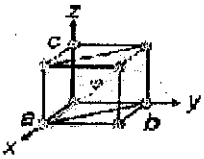
(a) similar electronegativity
(b) similar valence
(c) similar crystal structure
(d) all above

5- As temperature decreases, the fraction of the total number of atoms that are capable of diffusive motion
- (a) Decreases
- (b) Increases
- (c) Both A and B
- (d) Monotone

6- What is the predominant type of bonding for (Al_2O_3)?
- (a) Ionic
- (b) Hydrogen
- (c) Covalent
- (d) Metallic

metal-nonmetal

7- The crystallographic plane shown can be represented by
- (a) (100)
- (b) (001)
- (c) (101)
- (d) (110)



8- The coordination number for atoms in FCC crystalline structure is:
- (a) 4
- (b) 8
- (c) 12
- (d) 16

9- If grain are randomely oriented, over all properties are
- (a) not directional
- (b) directional
- (c) anisotropic
- (d) Both A and C

the structure is isotropic!

isotropic = randomly oriented!

Question is asking for properties!

10- The equilibrium concentration of a point defect, such as a vacancy, is proportional to:
- a) exp(-NA).
- b) the crystal structure.
- c) exp(-Qv).
- d) the energy required for its formation.

$$N = N_A \exp\left(-\frac{Q_v}{kT}\right)$$

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Exercise 2 (10 points)

An x-ray diffractometer recorder chart for an element that has either the BCC or the FCC crystal structure showed diffraction peaks at the following 2θ angles: 38.60°, 55.71°, 69.70°, 82.55°, 95.00°, and 107.67°. Wavelength λ of the incoming radiation was 0.15405 nm.

$\boxed{n=1}$

(a) Determine the crystal structure of the element

(b) Determine the lattice constant of the element

(c) Identify the element

$\frac{\sin^2\frac{38.6°}{2}}{\sin^2\frac{55.71°}{2}} = \frac{\sin^2\theta_A}{\sin^2\theta_B} = \frac{(h^2+k^2+l^2)_A}{(h^2+k^2+l^2)_B} = 0.5$

100
110
111
200
210
211

a) Since the resultant angle is odd, the structure is BCC.

$2\theta = 38.60°$
$\theta = 19.3°$

Solution # 2

b) $d = \frac{n\lambda}{2\sin\theta_c} = \frac{(1)(0.15405\,nm)}{2\sin(19.30°)} = 0.23305\,nm$

lattice constant = a!

$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$ Since the structure is BCC, $h^2+k^2+l^2$ must be even!

$0.23305\,nm = \frac{a}{\sqrt{(1)^2+(1)^2+(0)^2}}$

$0.23305\,nm = \frac{a}{\sqrt{2}}$ → even!

$(0.23305)(\sqrt{2}) = a$

$\boxed{0.32958\,nm = a}$

c) since the crystal structure is BCC, use $a_{BCC} = \frac{4R}{\sqrt{3}}$ to solve for R.

$\frac{(0.32958\,nm)(\sqrt{3})}{4} = R$

$\boxed{0.1430\,nm = R}$

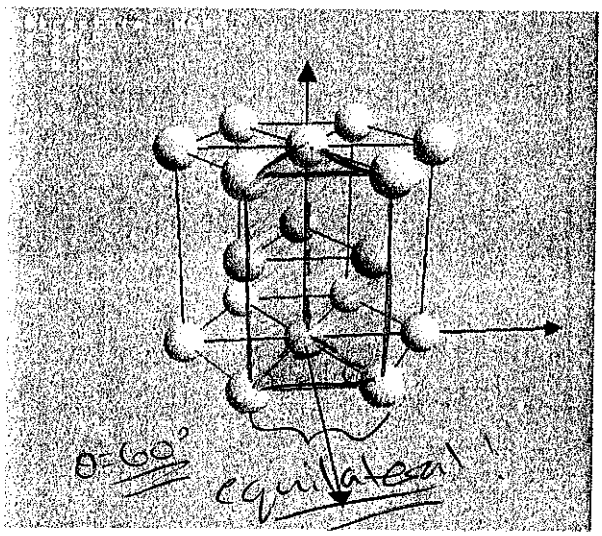
Looking at the chart provided, Tantalum has a BCC structure and an atomic radius of 0.1430 nm.

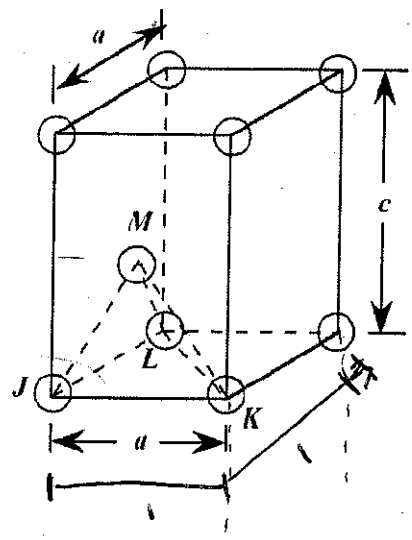
∴ the element is tantalum!

Problem # 3 (5 marks)

Show that the ideal c/a ratio for HCP is 1.633. A sketch of one-third of an HCP unit cell and the hexagonal structure are shown below.

$\frac{\sin 90^\circ}{1} = \frac{\sin 90^\circ}{c}$





Solution#3

Assuming that we are working with a unit cell HCP structure, we know that length 'a' (the shortest side) is not equal to length 'c' (the longest side).

$a = b \neq c$; $\alpha = \beta = \gamma = 90^\circ$

The atomic packing factor of HCP is 0.74

~~[illegible]~~

$c/a = 1.633$

$a = 1$ in a unit cell

$1/0.74 = 1.351$

1.633 is the ideal value, but actual values vary slightly.

0

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APF = $\frac{\text{volume of atoms}}{\text{volume of cell}}$ $\qquad a = 2R$, $c = 2R$

$$0.74 = \frac{\frac{4}{3}\pi r^3 (6)}{3a^2 \sin 60^\circ c}$$

$$\boxed{V_{HCP} = 3a^2 \sin 60^\circ c}$$

$$1.923 = \frac{24\pi r^3}{3a^2 c}$$

$$5.768 = \frac{24\pi r^3}{a^2 c}$$

$$5.768 a^2 c = 24\pi r^3$$

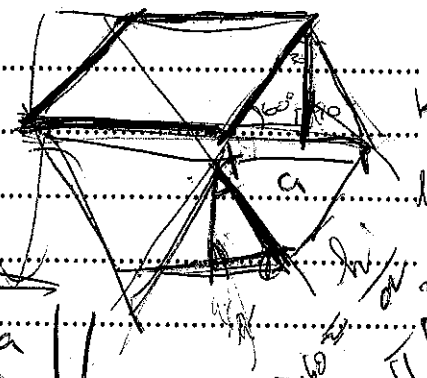
$$a^2 c = 13.07 r^3$$

$$a^2 c = 13.07 \left(\frac{1}{2}a\right)^3$$

$$a^2 c = \frac{13.07 a^3}{8}$$

$$\frac{a^2 c}{a^3} = 1.633$$

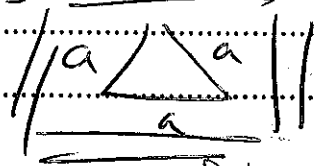
$$\boxed{\frac{c}{a} = 1.633}$$



lozenge

$h = 2a$

$h = ?$ $\sin 60^\circ = \frac{h}{a}$



equilateral

$6 \times \frac{1}{2} a h$

$$0.74 = \frac{6\left(\frac{4}{3}\pi r^3\right)}{6 \times \frac{1}{2} a \sin 60^\circ \times c}$$

base depth

base A 1 tri

Area $\times$ C

multiply $\frac{V_S}{V_U}$ by c because this is volume of triangular prism.

$\sin 60^\circ = \frac{h}{a}$

$h = (\sin 60^\circ) a$

must put in terms of a!

$\frac{V_S}{V_U} = 6\left(\frac{4}{3}\pi r^3\right) / 6 \times \frac{1}{2} a h \sin 60^\circ$

6 equilateral $\triangle$'s area of $\triangle$

$h = a \sin 60^\circ$